Hybrid CNN-BiLSTM Model for Power Quality Disturbance Classification

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***Abstract*—With the evolution of intelligent grid tech, the need for monitoring power quality has markedly risen, essential for the continuous functioning of the power grid. Addressing this matter, this research proposes a combined approach combining Convolutional Neural Networks (CNN) with Bidirectional Long Short-Term Memory (BiLSTM) networks. The model effectively catches spatial or temporary aspects of power quality disruptions using CNN and employs BiLSTM to document long-term or temporal aspects, thus creating a profound correlation between input and output sequences. By substantially decreasing reliance on conventional physical attributes, this technique markedly enhances the precision in detecting power quality anomalies. Crucially, this technique, independent of particular original samples and existing information, holds widespread applicability and is applicable in identifying patterns in various other intricate systems. Research on experiments indicates that, as opposed to conventional machine learning frameworks, deep learning approaches are more effective in detecting disruptions in power quality. Our hybrid model excels in identifying power quality disturbance (PQD) and produces highly accurate recognition outcomes by autonomously analyzing central features of the disturbance indications.**

Keywords—power quality disturbance, bidirectional Long Short-Term Memory, convolutional neural network, deep learning

I. INTRODUCTION

With the widespread use of intermittent and renewable resources in the power system, microgrids are facing various operational disturbances. There are various interference factors that affect power quality, including but not limited to unstable system loads and equipment failures. These issues may cause serious damage to the quality of electricity [1]. Therefore, finding an effective method for managing and monitoring PQDs in microgrid environments has gradually been recognized as a highly anticipated and important topic in the field of power system research [2]. In the power system, numerous renewable energy sources may cause various fluctuations, which are likely to lead to serious problems. Therefore, adopting appropriate coping strategies is crucial. Due to the unique performance of different power sources, this comprehensive strategy may increase the probability of changes in the power grid. Over time, the large-scale introduction of renewable energy may lead to disturbances in power quality, further posing a series of problems for the power grid system. Therefore, seeking efficient solutions to the difficulties in the suggestions is particularly crucial [3].

This study considers 10 types of single disturbance signals, including normal voltage, swell, sag, harmonics, flicker, interruption, impulsive transient, oscillatory transient, periodic notch, and spike. The single PQDs waveforms are shown in Fig. 1:

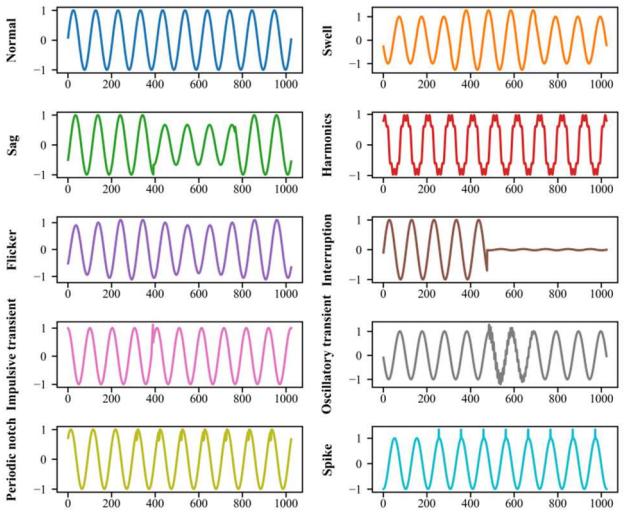


Fig. 1. Power quality single disturbance waveform.

Signal-to-Noise Ratio (SNR) is an indicator that measures the ratio of a signal to noise, measured in decibels (dB), and describes the relative strength of the signal to noise. In theory, it is necessary to add noise ranging from 70 dB to 20 dB to enhance the authenticity of the data and test the model's noise resistance and generalization ability. As shown in (1):

$$SNR(dB) = 10log_{10}\left(\frac{P_{signal}}{P_{noise}}\right) \quad (1)$$

P_{signal} is the signal power, and P_{noise} is the noise power.

The detection and recognition of PQD usually consists of two steps: feature extraction and feature classification. Feature

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extraction utilizes digital signal processing (DSP) techniques such as discrete Fourier transform (DFT), short-time Fourier transform (STFT) [4], wavelet transform (WT) [5], S-transform (ST) [6], Hilbert-Huang transform (HHT) [7], singular value decomposition (SVD) [8], Kalman filter (KF), etc; In terms of feature classification, traditional machine learning methods such as decision trees (DT) [9], support vector machines (SVM) [10], K-nearest neighbors (KNN), random forests (RF) [11], XGBoost, Lightgbm, and deep learning methods such as LSTM, CNN, and Transformer.

While conventional digital signal processing has evolved, it faces certain intrinsic limitations: the STFT window function's structure and dimensions remain unchanged, meaning it limits the temporal resolution and frequency segmentation for both static and sudden signal phases, and falls short in adaptability, being only apt for stationary signal analysis [12]; WT, as constrained by the Heisenberg uncertainty principle, is barred from achieving both high temporal and frequency resolutions. This also shows responsiveness to fleeting high-frequency signals like noise [13]. Consequently, enhancing the precision of detection requires efficient noise reduction preprocessing in the analysis of PQD signals. Although the SVD's time-domain analysis technique is devoid of the signal's frequency domain features, it frequently involves integration with various algorithms in detecting PQD signals [14].

As for feature categorization, hidden layers of LSTM usually include many more parameters and models are too complicated, plus insufficient or overly complex training data indeed affects the merit of the model [15]. As for the side of the inference issue of time criteria and perspectives, CNN tends to have its view constricted, and also time-sensitive features are difficult to obtain in long-length series data. Concerning feature sorting, LSTM architecture can be classified as finely detailed since a wide range of parameters is usually required. Such a detailed modeling practice is only deployable when dealing with complexities and extended timeline dependencies in the training data [16]. On the contrary, the CNN technique works better for events that have narrower viewpoints, for problems in time constraints, and in handling rigorous timeline series [17]. For Transformer, the major distinction is found in its proficiency in managing far-off dependencies that even CNN or LSTM approaches have difficulty with, either the convolutional or the recurrent neural network approach. With self-attention, Transformer models have their own associated complications that come along when dealing with proportional strings that have so long length that no approach including Transformer models will be able to handle them [18].

Given the challenge of integrating both local and global characteristics of input data into one model at once, this study focuses on enhancing recognition precision by merging CNN's local feature extraction with BiLSTM's global feature extraction feature retrieval. Results from the simulation studies reveal that the identification algorithm introduced here is capable of rapidly and precisely fulfilling the power quality recognition task, amidst significant noise, substantial data, and unidentified elements, highlighting its possible importance in engineering contexts.

The CNN is a feedforward neural network that includes one or more convolutional layers and a fully connected top layer. This method is especially adept at handling data featuring distinct hierarchical structures, like images, due to its proficiency in uncovering local data attributes via related weights and integration layers.

A Bidirectional Long Short Term Memory Network (BiLSTM) integrates both forward and reverse Long Short Term Memory (LSTM) components. LSTM, a unique form of recurrent neural network (RNN), is engineered to process and forecast events occurring at different time intervals in time series data. Owing to its distinctive architecture, LSTM proves highly effective in identifying long-term interdependencies, namely overarching characteristics. Fig. 2 displays the configuration of the CNN-BiLSTM.

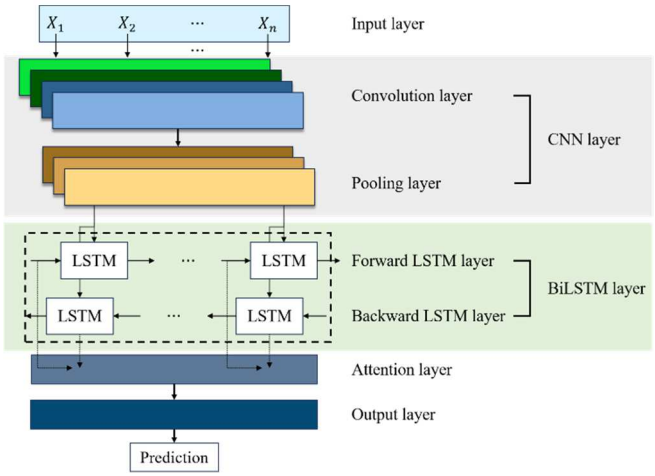


Fig. 2. Schematic diagram of CNN-BiLSTM.

II. Methodology

For the credibility of our suggested model, it's essential to have effective PQDs data, encompassing a range of disturbances. The IEEE Std1159 outlines various types of PQDs and their differing parameters [19]. Based on a mathematical framework, this study generated 2 completely independent datasets with the same parameters in Python, each containing 10 different synthesized PQD signals. C1 to C10 correspond to normal signal, swell, sag, harmonics, flicker, interruption, impulsive transient, oscillatory transient, periodic notch, and spike. In every type of disturbance, a thousand waveforms are produced through the random alteration of parameter values. The device captures the signal of 10 cycles in every PQD waveform. The fundamental frequency stands at 50 Hz, while the sampling rates are 5120 Hz. Under the specified conditions, every waveform captures data at 0.2s sampling period, encompassing 1024 data points.

With a high signal-to-noise ratio, the signal's strength significantly surpasses that of the noise, diminishing the overall effect of noise on the signal. Signal alterations might be subtle, resulting in minimal variation to the original signal after introducing a 50 dB noise ratio; conversely, the signal is significantly disrupted and becomes undetectable under the influence of a 10 dB white noise level [20]. This study incorporated signal-to-noise ratios of 20, 30, and 40 dB into the test dataset.

There are two datasets generated for the training and testing process. The training dataset contributes 70% for training and 30% for validating which will feed into the model to learn for the classification process. The completely new dataset will be used as testing data to evaluate this CNN-BiLSTM model prediction performance. Where training dataset consists of 10,000 data with 1000 data for each

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classified single signal. Similar to the testing dataset consists of 10,000 data with 1000 data for the same classified single signal. The first dataset is fed into a machine learning and deep learning model for training, and then the second dataset is fed into a training model. By comparing the recognition accuracy, it is verified that the method proposed in this paper has good performance in PQD recognition and classification. The process is shown in Fig. 3.

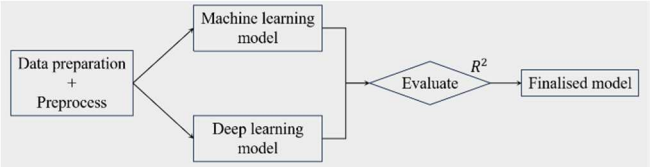


Fig. 3. Schematic diagram of this research

The technique exhibits a strong resilience in identifying and classifying PQD, utilizing the capabilities of both CNN and BiLSTM networks. This varied dataset facilitates extensive learning and assessment stages, enhancing the model's capacity to detect and accurately categorize various PQDs. This showcases the efficacy of the integrated CNN-BiLSTM framework in managing intricate PQD scenarios.

TABLE I. displays the hardware models and their software versions. This computer configuration, featuring an AMD Ryzen 9 5900HS with 8 cores and 16 threads and an RTX 3060 with 4GB of memory, along with 16GB DDR4 RAM, provides robust support for deep learning model training and inference. The Ryzen CPU's multi-threading capabilities significantly enhance parallel task processing, accelerating model training. The RTX 3060 GPU delivers substantial computational power and sufficient memory, ensuring the efficient and smooth execution of deep learning models.

TABLE I. MODELS OPERATING ENVIRONMENT

Hard/Soft-ware	Version
OS	Win10 64bit
CPU	AMD Ryzen 9 5900HS
GPU	RTX 3060
RAM	DDR4 16GB*1
Python	3.9.7
Keras	2.15.0
Tensorflow	2.15.0
Numpy	1.26.4

The aforementioned CNN-BiLSTM model is executed using Keras in Tensorflow, with epochs of 50. In response to the issue of overfitting during the training of the model, 50% of dropouts in each layer of the deep learning model were utilized. Besides, the model uses a batch size of 32 with a learning rate of 0.001 and uses Adam optimizer to improve the accuracy of the model by controlling the reduction of total loss when training using a training dataset.

III. SIMULATION RESULT AND ANALYSIS

Fig. 4 illustrates the comparative analysis of classification precision between this network and conventional classification neural networks in varied noise conditions:

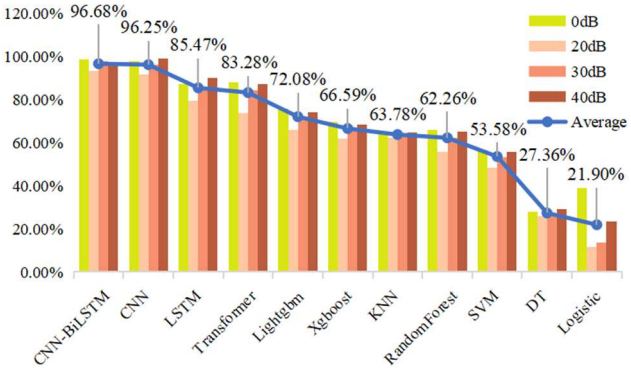


Fig. 4. Comparison of accuracy between the method uesd in this paper

Each of the four deep learning models namely Transformer, LSTM, CNN, and CNN-BiLSTM surpasses 80% in accuracy and outperforms conventional machine learning models, suggesting a more advanced capability in signal detection. The CNN-BiLSTM model introduced in this research demonstrates the greatest accuracy in recognition, achieving an average accuracy of 96.68% in classifying single PQD signals of four different signals to noise ratios of 0 dB, 20 dB, 30 dB, and 40 dB. Furthermore, noise of 20 dB exhibits potent interference, leading to a notable diminution in the recognition accuracy of each model under this noise level, particularly in Logistic models.

The confusion matrix in Fig. 5 clearly shows the predicted results for CNN-BiLSTM models with R^2 is 0.9863 in 0 dB, which means CNN-BiLSTM has excellent classification performance in signal recognition.

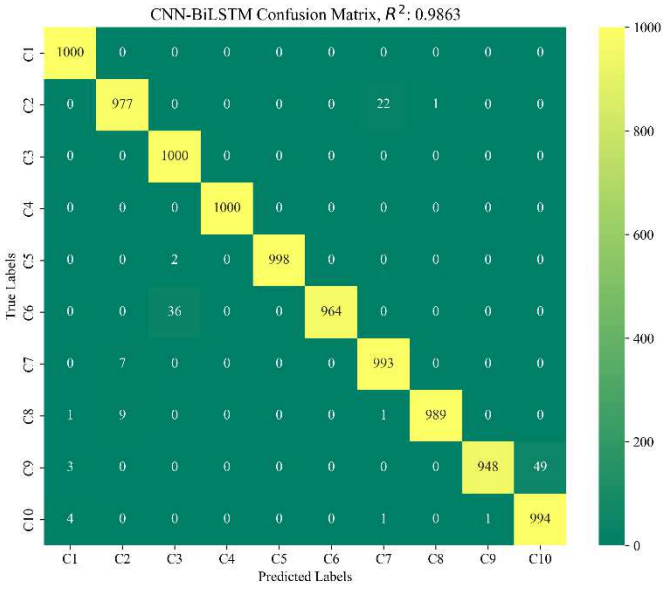


Fig. 5. The confusion matrix of CNN-BiLSTM without noise

The CNN-BiLSTM framework demonstrates difficulties in precisely detecting Swell, Interruption, and Periodic Notch disruption. On the basis of Swell, 977 out of 1000 signals turned out to be accurate, whereas 22 were misclassified as Impulsive Transients and the rest were Oscillatory Transients. The system precisely distinguished 964 out of 1000 Interruption signals, mislabeling the rest as Sag, highlighting the difficulties in precisely identifying such interruptions from the model. In relation to Periodic Notch (C9), 3 out of 1000

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signals were incorrectly identified as Normal (C1) and 49 out of 1000 signals were incorrectly identified as Spike (C10).

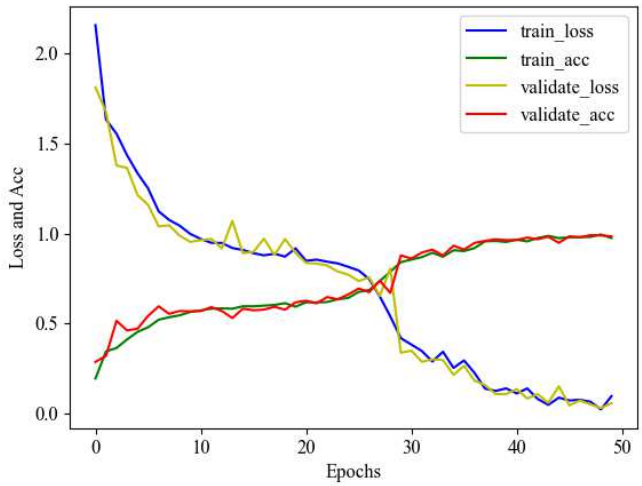


Fig. 6. The loss and acc rate of CNN-BiLSTM without noise

Fig. 6 shows a plot displaying CNN-BiLSTM's accuracy and loss functions in training and validation sets. The levels of training and validation accuracy for CNN-BiLSTM improved and accordingly decreased the levels of training and validation loss. Finally, the training accuracy increased up to 98.3%, and the validation increased to 98.4%. Also, improvement in training loss (2.35 to 0.11) and validation loss (1.77 to 0.09) was discovered. The results give the parameters of the framework a high score, and the training particularly wins the qualification for the data. Further analysis by looking into each individual single PQD with different SNRs using the CNN-BiLSTM model will be discussed hereinafter.

Following are the four tables that extensively analyze the performance of the CNN-BiLSTM model against SNRs of 0 dB, 20 dB, 30 dB, and 40 dB. By defining this as a model's ability to identify various disturbance signals and this ability specifically, the assessments are focused. Tables shows precision, recall, and F1-score as the primary performance measures for the analysis.

TABLE II. CNN-BiLSTM PERFORMANCE WITHOUT NOISE

SNR	Disturbance term	Evaluation index		
		Precision	Recall	F1-score
0 dB	C1	99.21%	100.00%	99.60%
	C2	98.39%	97.70%	98.04%
	C3	96.34%	100.00%	98.14%
	C4	100.00%	100.00%	100.00%
	C5	100.00%	99.80%	99.90%
	C6	100.00%	96.40%	98.17%
	C7	97.64%	99.30%	98.46%
	C8	99.90%	98.90%	99.40%
	C9	99.89%	94.80%	97.28%
	C10	95.30%	99.40%	97.31%
	Average	98.67%	98.63%	98.63%

TABLE III. CNN-BiLSTM PERFORMANCE WITH SNR OF 20 DB

SNR	Disturbance term	Evaluation index		
		Precision	Recall	F1-score
20 dB	C1	68.39%	93.70%	79.07%
	C2	99.11%	99.90%	99.50%
	C3	99.70%	99.30%	99.50%
	C4	100.00%	100.00%	100.00%
	C5	99.90%	99.20%	99.55%
	C6	99.50%	99.70%	99.60%
	C7	99.90%	98.20%	99.04%
	C8	96.04%	94.60%	95.31%
	C9	97.05%	62.40%	75.96%
	C10	85.98%	87.70%	86.83%
	Average	94.56%	93.47%	93.44%

TABLE IV. CNN-BiLSTM PERFORMANCE WITH SNR OF 30 DB

SNR	Disturbance term	Evaluation index		
		Precision	Recall	F1-score
30 dB	C1	96.16%	95.20%	95.68%
	C2	99.01%	99.90%	99.45%
	C3	99.90%	97.50%	98.68%
	C4	100.00%	100.00%	100.00%
	C5	99.80%	100.00%	99.90%
	C6	97.56%	99.90%	98.72%
	C7	99.19%	98.20%	98.69%
	C8	99.79%	96.80%	98.27%
	C9	88.37%	97.30%	92.62%
	C10	99.79%	93.60%	96.59%
	Average	97.96%	97.84%	97.86%

TABLE V. CNN-BiLSTM PERFORMANCE WITH SNR OF 40 DB

SNR	Disturbance term	Evaluation index		
		Precision	Recall	F1-score
40 dB	C1	86.88%	100.00%	92.98%
	C2	99.39%	98.10%	98.74%
	C3	100.00%	97.20%	98.58%
	C4	100.00%	100.00%	100.00%
	C5	100.00%	100.00%	100.00%
	C6	97.28%	100.00%	98.62%
	C7	99.90%	95.70%	97.75%
	C8	96.54%	94.90%	95.71%
	C9	93.87%	87.30%	90.47%
	C10	95.56%	94.70%	95.13%
	Average	96.94%	96.79%	96.80%

TABLE II. shows the performance of the model without noise. The C1, C3, and C4 were correctly identified, while the recognition rates of the other signals were all above 94%. TABLE III. shows the anti-interference ability of the model in 20 dB. Although the accuracy of C9 and C10 is relatively low, at 62.40% and 87.70%, respectively, the overall average accuracy reached 93.47%, which is satisfactory. In 30 dB, the result in TABLE IV. surprisingly found that the model's recognition rate for C9 was significantly higher than other SNRs. TABLE V. shows the performance of the model in 40dB, with 4 signals being perfectly recognized, demonstrating the model's excellent performance in high SNR.

From the above classification results, it is noticeable that the classification accuracy without noise interference can reach nearly 99%. Besides, it is interesting to find out that the model able to constantly predict C4 with 100% accuracy for every noise. The average accuracy of single PQDs classification performance for four different SNRs of 0, 20, 30 and 40 dB is 98.63%, 93.47%, 97.84%, and 96.79% respectively. The 20 dB SNR has the lowest accuracy of 93.47% compared to 0, 30, and 40 dB. This is aligned with many other

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researchers showing that with 20 dB interference, the weak oscillation signal is mostly covered by noise.

Precision and F1-score have the same trend at different SNRs as Recall. The consistent trends of Precision and Recall in different SNRs indicate that the model can maintain stable performance in various environments. This situation indicates that the model can make reasonable trade-offs when processing data, ensuring the stability and consistency of overall performance and performing.

By testing the model's performance in different SNRs and recording the changes in Precision, Recall, and F1-score, the model's effectiveness in various SNRs can be evaluated. This evaluation method helps to demonstrate the robustness of the model. In addition, more attention should be paid to the challenges of deploying such models in real-world smart grid environments. Specific challenges may include processing large-scale real-time data, dealing with diverse PQDs, and optimizing the computational efficiency of models.

Implementing real-time monitoring for PQD identification is crucial. This model requires about 20 minutes of offline training, after which it can monitor PQDs in real-time online. To maintain system efficiency and accuracy, the monitoring system can be designed to retrain newly collected disturbance data at regular intervals and update the model accordingly. This approach leverages big data on PQDs to continually enhance and optimize the monitoring system, ensuring high performance and reliability in dynamic environments.

IV. Conclusion

The model proposed illustrates the possibility of developing hybrid deep-learning architectures for efficiently identifying and classifying PQD in a way that allows for the effective operation of power systems. This study demonstrates the progress of deep learning as applied to an area inclined to signal processing and machine learning techniques. A comprehensive comparison using different machine learning (LightGBM, XGBoost, KNN, Random Forest, Support Vector Machine, Decision Tree, and Logistic) and deep learning (LSTM, CNN, Transformer, and CNN-BiLSTM) techniques were studied for the classification of single-signal PQDs. Through running the analysis by using similar hyperparameters (such as batch size, epoch, and learning rate) to control the true comparison, it has been demonstrated that the deep learning model incorporating CNN with BiLSTM networks outperforms traditional machine learning and other deep learning models in terms of prediction accuracy which evaluated using R^2 values. Specifically, the CNN-BiLSTM model has shown exceptional capability in classifying 10 single PQD with R^2 values of 0.9668, which is 24.6% more accurate than the most performing machine learning model namely LightGBM with R^2 values of 0.7208.

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